ANTHROPOMORPHISM, SELF-DISCLOSURE AND WELL-BEING AMONG GENERATION Z USERS OF CONVERSATIONAL AGENTS IN MALAYSIA

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A thesis submitted in fulfilment of the requirements for the award of the degree of Doctor of Philosophy

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2026

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ABSTRACT

Conversational agents are now part of daily life for many young adults. This study examined how anthropomorphism relates to self-disclosure and psychological well-being among Generation Z users in Malaysia. A total of 412 respondents completed an online survey. Anthropomorphism predicted greater self-disclosure ($\beta = .38$, $p < .001$), and self-disclosure was associated with higher well-being ($\beta = .21$, $p < .01$). Self-disclosure partially mediated the relationship between anthropomorphism and well-being.

CHAPTER 1 INTRODUCTION

1.1 Introduction

Conversational agents such as ChatGPT and Replika have moved from novelty to daily companions for many young people. Generation Z users talk to these systems about their studies, their worries and their relationships. This chapter introduces the study, its background, the problem it addresses and its objectives.

1.2 Background of the Study

Anthropomorphism is the tendency to attribute human characteristics, motivations and emotions to non-human agents. Users who perceive a chatbot as human-like often treat it as a social partner rather than a tool. Research on social presence suggests that human-like cues encourage people to share personal information. In Malaysia, more than 80% of young adults use at least one conversational agent each week. Self-disclosure is known to support well-being when it leads to feeling understood and supported.

1.3 Problem Statement

Despite rapid adoption, little is known about how anthropomorphism shapes what young users disclose to conversational agents. Most studies have been conducted in Western samples, and evidence from Southeast Asia remains scarce. Existing findings on chatbot use and well-being are also inconsistent, with some studies reporting benefits and others reporting increased loneliness. Without this evidence, designers and counsellors lack guidance on healthy use.

1.4 Research Questions

This study seeks to answer the following research questions: RQ1: What is the level of anthropomorphism perceived by Generation Z users of conversational agents? RQ2: Is anthropomorphism associated with self-disclosure to conversational agents? RQ3: Does self-disclosure mediate the relationship between anthropomorphism and psychological well-being?

1.5 Research Objectives

The objectives of this study are: (i) to identify the level of anthropomorphism perceived by Generation Z users; (ii) to examine the relationship between anthropomorphism and self-disclosure; and (iii) to determine the mediating role of self-disclosure in the relationship between anthropomorphism and psychological well-being.

1.6 Conceptual Framework

The framework is based on Social Penetration Theory and the Computers Are Social Actors paradigm. In this framework, anthropomorphism is the independent variable, psychological well-being is the dependent variable, and self-disclosure is the mediating variable. Loneliness is examined as the moderating variable.

1.7 Scope of the Study

This study focuses on Generation Z university students aged 18 to 27 years in Peninsular Malaysia. Data were collected between March and June 2025 through an online questionnaire. The study is limited to text-based conversational agents and does not include voice assistants.

1.8 Significance of the Study

Theoretically, this study extends Social Penetration Theory to human-AI interaction by testing self-disclosure as a mechanism. Practically, the findings can inform designers of conversational agents about features that encourage healthy disclosure. The results will also help university counsellors understand how students use chatbots for emotional support.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 Research Design

This study used a quantitative cross-sectional survey design. Data were analysed with partial least squares structural equation modelling.

3.2 Phase 1: Instrument Development

The questionnaire was adapted from validated scales and reviewed by three experts in psychology and human-computer interaction.

3.3 Phase 2: Pilot Study

A pilot study with 40 respondents confirmed the reliability of all scales, with Cronbach alpha values above .80.

3.4 Phase 3: Main Data Collection

A total of 412 respondents were recruited through purposive sampling from five public universities.

3.5 Phase 4: Data Analysis

The measurement and structural models were assessed using SmartPLS 4, and mediation was tested with bootstrapping of 5,000 samples.

CHAPTER 4 RESULTS

4.1 Respondent Profile

Of the 412 respondents, 63% were female and the mean age was 21.4 years (SD = 2.1). Most respondents used ChatGPT, followed by Character.AI and Replika.

4.2 Level of Anthropomorphism

The mean level of anthropomorphism was moderate (M = 3.42, SD = 0.71). Respondents most often described their chatbot as friendly and understanding.

4.3 Structural Model

Anthropomorphism significantly predicted self-disclosure (β = .38, p < .001). Self-disclosure was associated with higher psychological well-being (β = .21, p < .01). The indirect effect of anthropomorphism on well-being through self-disclosure was significant (β = .08, 95% CI [.03, .14]), indicating partial mediation. Loneliness strengthened the link between anthropomorphism and self-disclosure.

CHAPTER 5 DISCUSSION AND CONCLUSION

5.1 Discussion

The findings show that users who see conversational agents as more human-like disclose more personal information. This pattern supports the Computers Are Social Actors paradigm. Disclosure in turn was linked to modest gains in well-being, which suggests that chatbots can offer a space to express feelings.

5.2 Implications

Theoretically, the results extend Social Penetration Theory by showing that disclosure to a non-human partner follows similar processes. Practically, designers should balance human-like features with clear reminders that the agent is not a human. University counselling services could use these findings to guide students on healthy chatbot use.

5.3 Recommendations

Universities should include digital well-being and AI literacy in student orientation programmes. Developers should provide transparent information on how disclosed data are stored and used. Counsellors should ask students about their chatbot use as part of routine assessment.

5.4 Limitations

The cross-sectional design prevents causal conclusions, and all measures were self-reported.

5.5 Suggestions for Future Research

Future research should use longitudinal designs to test whether disclosure to chatbots predicts later well-being. Future studies should also include older adults and voice-based agents. Experimental studies could manipulate human-like cues to test their causal effect on disclosure.

5.6 Conclusion

Anthropomorphism encourages young Malaysians to open up to conversational agents, and this openness is linked to better well-being. Healthy design and guidance can help young users benefit while staying safe.

REFERENCES

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